CAMPUS PLACEMENT MANAGEMENT SYSTEM

Project Overview

The Campus Placement Management System is a Python-based application designed to automate and manage the campus recruitment process in colleges and universities. The system helps manage students, companies, placement drives, eligibility criteria, interview schedules, job offers, and placement statistics.

This project simulates a real-world placement portal used by colleges and provides hands-on experience in Python programming, file handling, data management, and Object-Oriented Programming concepts.

Objectives

The system should be able to:

1. Manage student profiles.
2. Register recruiting companies.
3. Conduct placement drives.
4. Check student eligibility.
5. Schedule interviews.
6. Manage job offers.
7. Track placement statistics.
8. Generate placement reports.

Functional Modules

1. Student Management

Features

- Student Registration
- Academic Record Management
- Resume Information
- Skill Management
- Placement Status Tracking

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2. Company Management

Features

- Company Registration
- Job Profile Management
- Package Details
- Eligibility Criteria
- Company Information Management

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3. Placement Drive Management

Features

- Create Placement Drives
- Schedule Drive Dates
- Assign Companies
- Manage Registration

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4. Eligibility Management

Features

- CGPA Verification
- Backlog Verification
- Skill Matching
- Automatic Eligibility Checking

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5. Interview Management

Features

- Interview Scheduling
- Interview Panel Assignment
- Interview Status Tracking

- Selection Updates

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6. Job Offer Management

Features

- Offer Letter Management
- Package Tracking
- Offer Acceptance/Rejection
- Multiple Offer Handling

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7. Placement Statistics

Features

- Placement Percentage
- Highest Package
- Average Package
- Department-wise Placement Analysis

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8. Report Management

Features

- Student Placement Report
- Company Recruitment Report
- Placement Statistics Report
- Offer Summary Report

Class Design

Abstract Class

Person

Attributes

- person_id

- name
- email
- contact_number

Methods

- display_details()
- update_details()

Derived Classes

Student (Inherits Person)

Attributes

- student_id
- branch
- cgpa
- skills
- placement_status

Methods

- apply_for_drive()
- view_offers()
- update_skills()

PlacementOfficer (Inherits Person)

Attributes

- officer_id
- designation

Methods

- create_drive()
- generate_report()
- schedule_interview()

Other Classes

Company

Attributes

- company_id
- company_name
- package
- eligibility_cgpa
- job_role

Methods

- register_company()
- update_company()
- display_company()

PlacementDrive

Attributes

- drive_id
- company
- drive_date
- eligible_students

Methods

- create_drive()
- register_students()
- display_drive_details()

Interview

Attributes

- interview_id

- student
- company
- interview_date
- status

Methods

- schedule_interview()
- update_status()

JobOffer

Attributes

- offer_id
- student
- company
- package
- offer_status

Methods

- generate_offer()
- accept_offer()
- reject_offer()

PlacementReport

Attributes

- report_id
- generated_date

Methods

- generate_statistics()
- export_report()

CampusPlacementSystem

Attributes

- students
- companies
- drives
- interviews
- offers
- reports

Methods

- register_student()
- register_company()
- create_drive()
- check_eligibility()
- schedule_interviews()
- generate_reports()
- save_data()
- load_data()

Python Concepts to be Implemented

Basic Python

✓ Variables
✓ Data Types
✓ Operators
✓ Conditional Statements
✓ Loops

Functions

User Defined Functions

- check_eligibility()

- generate_offer()
- create_drive()

Recursive Function

- Search student records recursively.

Lambda Functions

- Sort students by CGPA.
- Sort companies by package.
- Sort placement records by salary.

Data Structures

List

Store students, companies, interviews, offers.

Dictionary

Store placement and company records.

Set

Store unique skills and technologies.

Tuple

Store fixed interview schedules.

Advanced Python Concepts

List Comprehension

Generate list of eligible students.

Decorators

Create logging decorators for:

- Student Registration
- Drive Creation
- Offer Generation
- Report Generation

Generators

Generate placement reports one record at a time.

Modules

Divide the project into multiple Python files.

OOP Concepts to be Implemented

Classes and Objects

Create objects for all classes.

Constructor

Use __init__() in every class.

Encapsulation

Protect student records and placement information.

Inheritance

Person → Student, PlacementOfficer

Polymorphism

Implement display_details() differently for Student, Company, and Placement Officer.

Abstraction

Person class should be abstract.

Composition

CampusPlacementSystem contains students, companies, drives, interviews, offers, and reports.

Static Methods

Eligibility and package calculation utilities.

Class Methods

Generate institution-wide placement statistics.

Magic Methods

Implement:

- __str__()

- __repr__()

File Handling

JSON Files

Store:

- Student Records
- Company Records
- Placement Drives
- Job Offers
- Interview Records

CSV Files

Generate:

- Placement Reports
- Company Reports
- Salary Analysis Reports

Exception Handling

Handle the following exceptions:

- Invalid Student ID
- Invalid Company ID
- Student Not Eligible
- Duplicate Registration
- Invalid CGPA Entry
- File Not Found
- Incorrect User Input

Modules to be Used

- datetime

- json
- csv
- os
- abc
- random

Menu Driven Interface

1. Student Management
2. Company Management
3. Placement Drive Management
4. Eligibility Checking
5. Interview Management
6. Job Offer Management
7. Placement Statistics
8. Report Generation
9. Save Data
10. Load Data
11. Exit